3 C#

```
int nextMark, sum, averageMark;
string markString;
sum = 0;
for (int mark = 1; mark <= 5; mark++)
{
    Console.WriteLine("Please enter a mark");
    markString = Console.ReadLine();
    nextMark = int.Parse(markString);
            sum = sum + nextMark;
      }

      averageMark = sum / 5;
      Console.WriteLine(averageMark);
```

a What is the purpose of the program?

b Why is `int.Parse` used in the line `nextMark = int.Parse(markString);`?

SKILLS PROBLEM SOLVING

ACTIVITY 8

Produce an algorithm that will print out the times tables (up to 12 times) for the numbers 2 to 12.

INDEFINITE ITERATION

An indefinite loop is used when the number of times a loop will need to be repeated is not known in advance. For example, if you want to give a user the option of playing a game as often as they want. Indefinite loops are repeated until a specified condition is reached.

Python

For indefinite iteration, Python uses the 'while' loop – something is done while a condition is met.

The following program asks a user to enter a number while the number is less than 20.

```
number = 1
while number <= 20:
    number = int(input('Please enter a number'))
print('You entered a number greater than 20')
```

As soon as the number is greater than 20, the program breaks out of the loop and prints a message for the user.

Java

For indefinite iteration, Java uses a 'while' loop – instructions are repeated while a condition is met.

The following program asks a user to enter a number while the number is less than 20.

```
import java.util.*;
class Main {
  public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);
    int number = 1;
    while(number <= 20) {
      System.out.print("Please enter a number: ")
      number = scanner.nextInt();
    }
    System.out.println("You entered a number greater than 20")
    scanner.close();
  }
}
```

C#

```
string numberString;
int number = 1;
while (number <=20)
{
    Console.WriteLine("Please enter a number");
    numberString = Console.ReadLine();
    number = int.Parse(numberString);
}

Console.WriteLine("You entered a number greater than 20");
```

As soon as the number is greater than 20, the program breaks out of the loop and prints a message for the user.

ACTIVITY 9

What do these two algorithms do? Implement them in the high-level programming language you are studying.

a Algorithm A

```
FOR index FROM 1 TO 10 DO
   SEND index * index * index TO DISPLAY
END FOR
```

b Algorithm B

```
SET counter TO 10
WHILE counter > 0 DO
```



▲ Calculators have a number of useful functions

SKILLS REASONING

SKILLS CRITICAL THINKING

SKILLS PROBLEM SOLVING

2 Develop an algorithm for a simple calculator and code it in a high-level language that:
   - **a** allows the user to choose from these options: addition, subtraction, division and multiplication
   - **b** prompts the user to input two numbers
   - **c** performs the calculation and displays the result
   - **d** offers the user the option of performing another calculation.

CHECKPOINT

Strengthen

S1 Why it is important to make your code easy to read?

S2 Outline the four techniques that a programmer should use to make code easy to read.

Challenge

C1 Revisit the programs you have already written. Do you still understand what they do and how they work? If not, try to improve their readability.

KEY POINT

Readability of code is important. You should always make your code easy to read.

How confident do you feel about your answers to these questions? If you're not sure you answered them well, look again at Table 2.5.

SUMMARY

- Using comments, descriptive names, indentation and white space makes code easier to read.
- Producing readable code makes it easier to understand what a program does and how it does it.